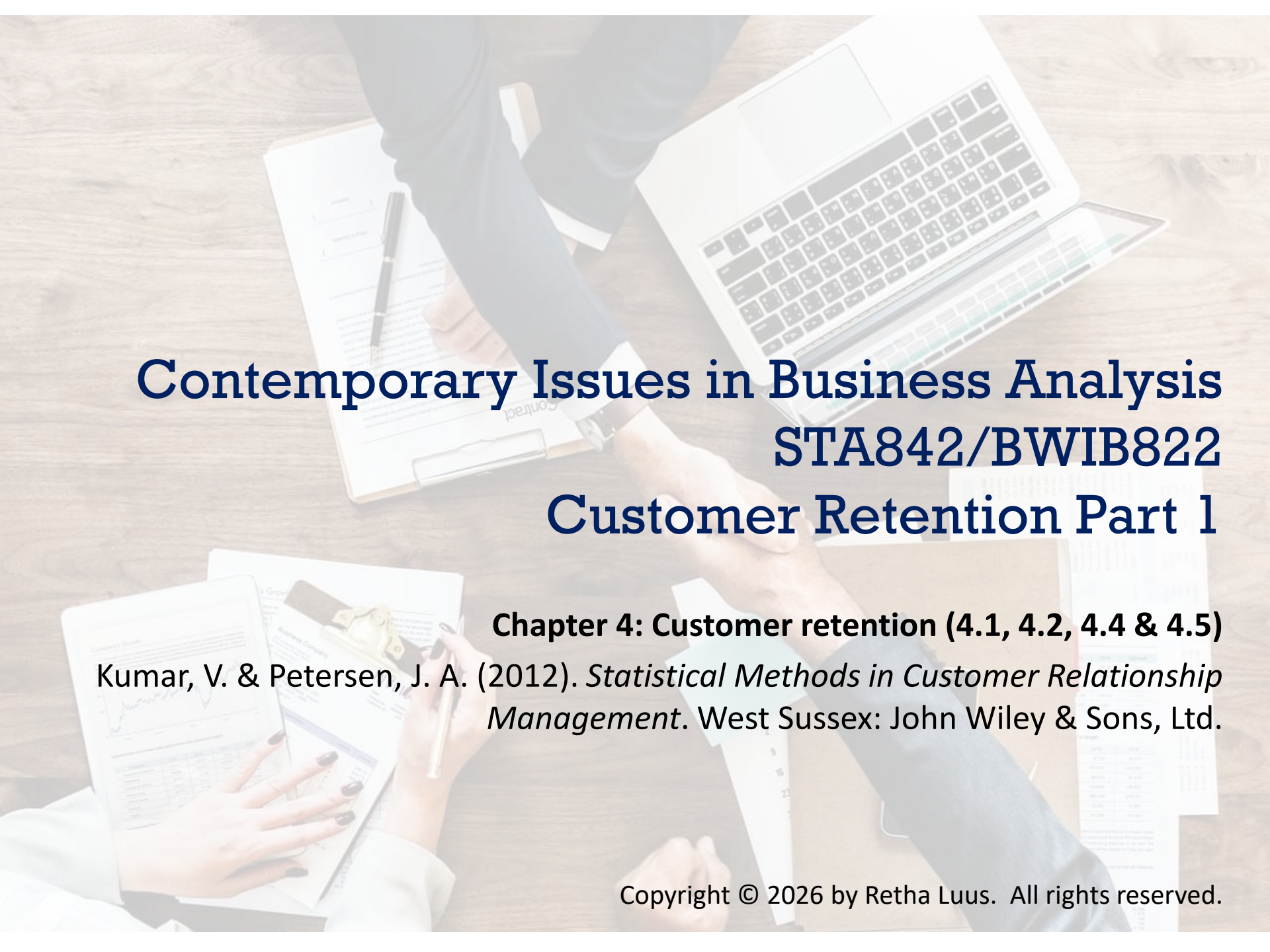
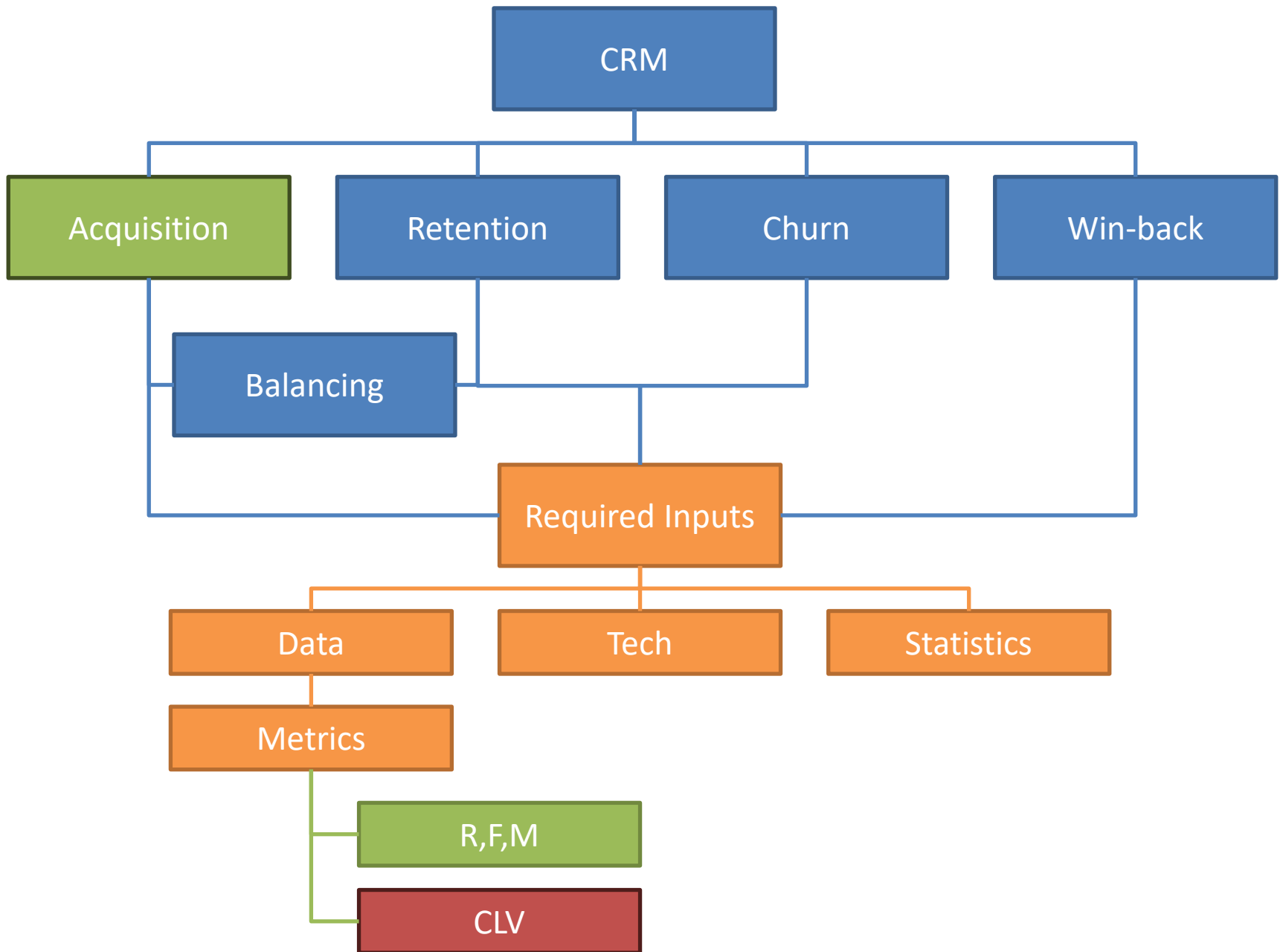


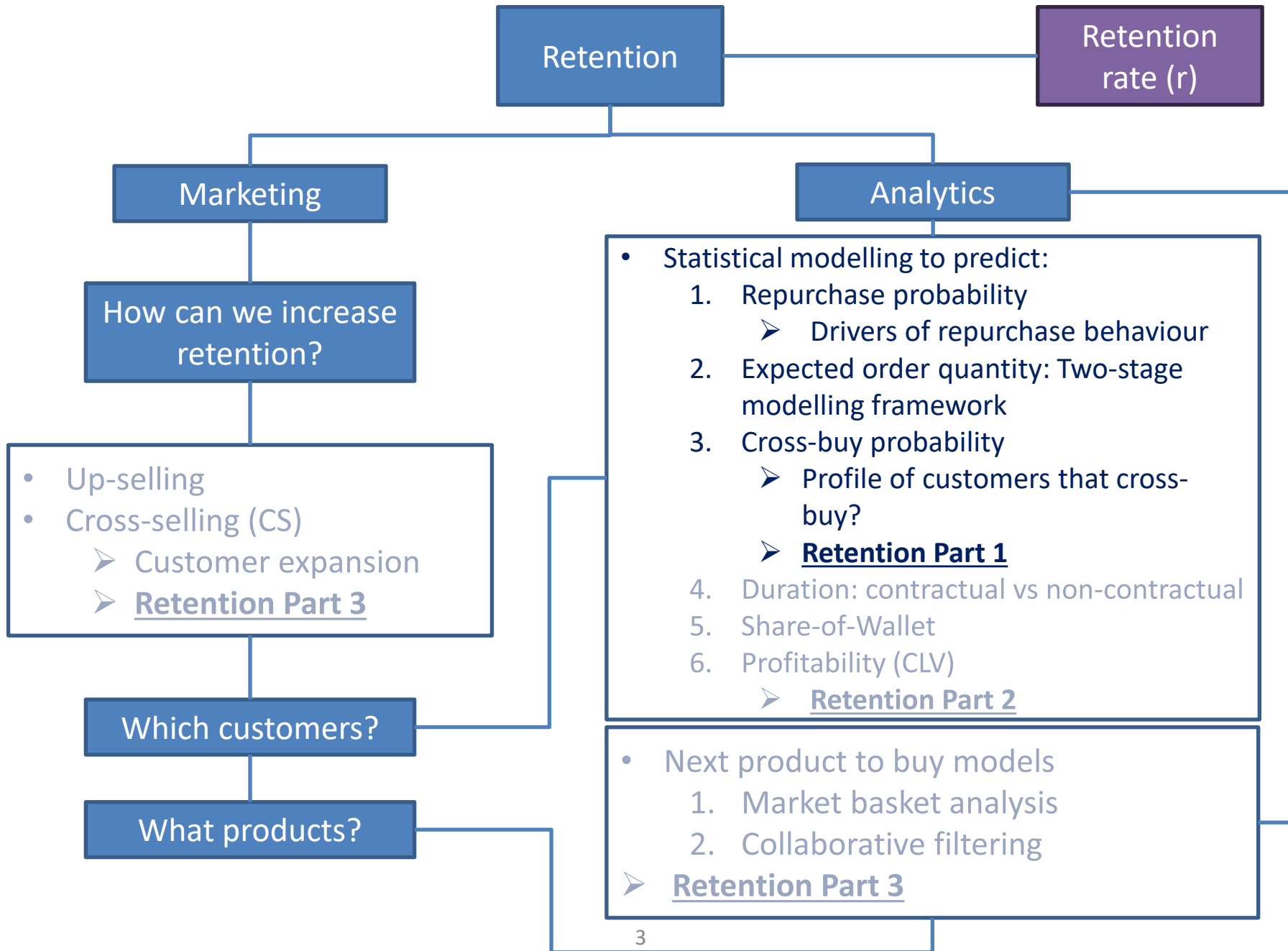


Contemporary Issues in Business Analysis STA842/BWIB822 Customer Retention Part 1

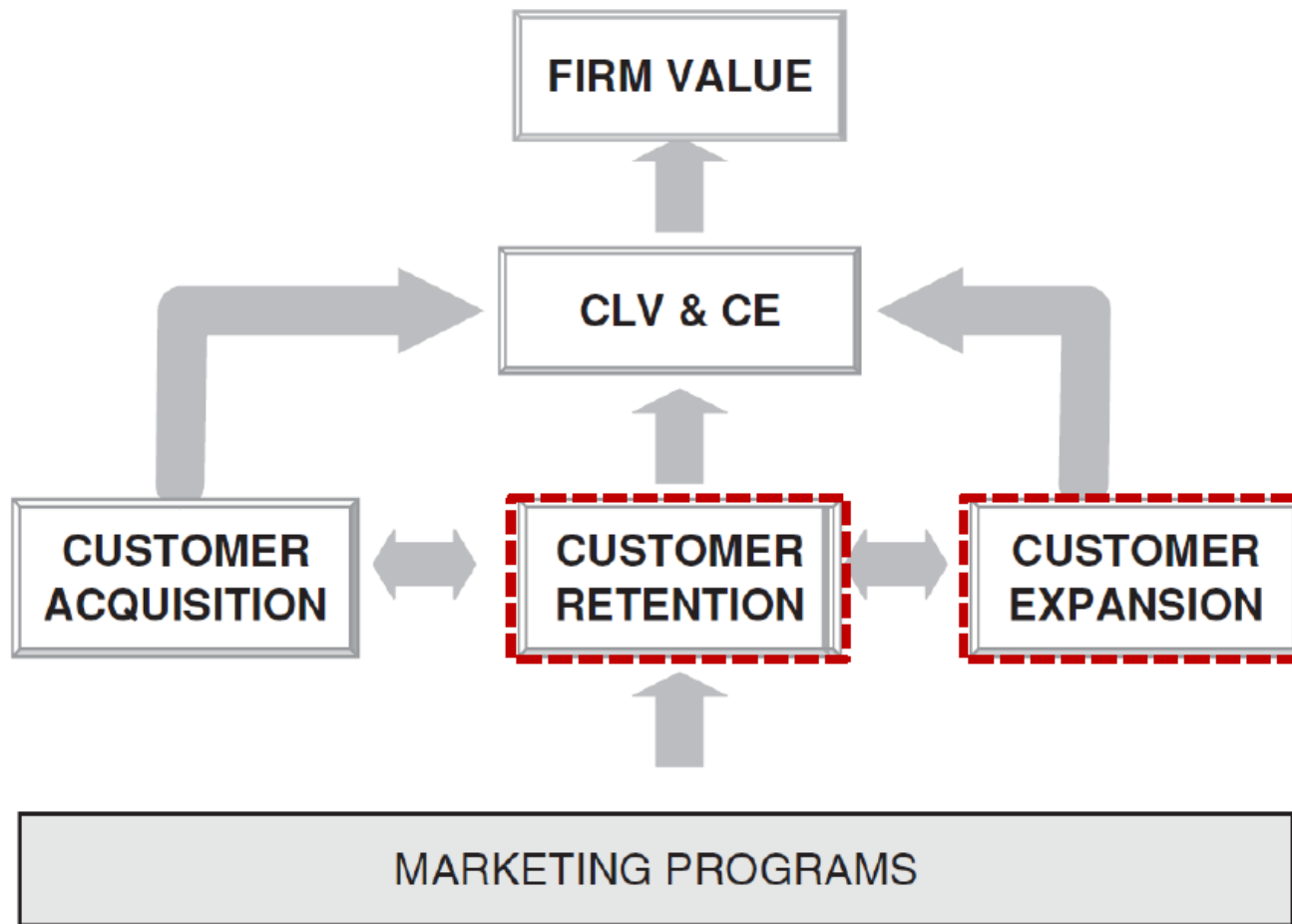
Chapter 4: Customer retention (4.1, 4.2, 4.4 & 4.5)

Kumar, V. & Petersen, J. A. (2012). *Statistical Methods in Customer Relationship Management*. West Sussex: John Wiley & Sons, Ltd.





CLV Conceptual Modelling Framework



Introduction



After acquisition, the focus shifts to customer retention



As before, retention strategies are used in contractual and non-contractual settings

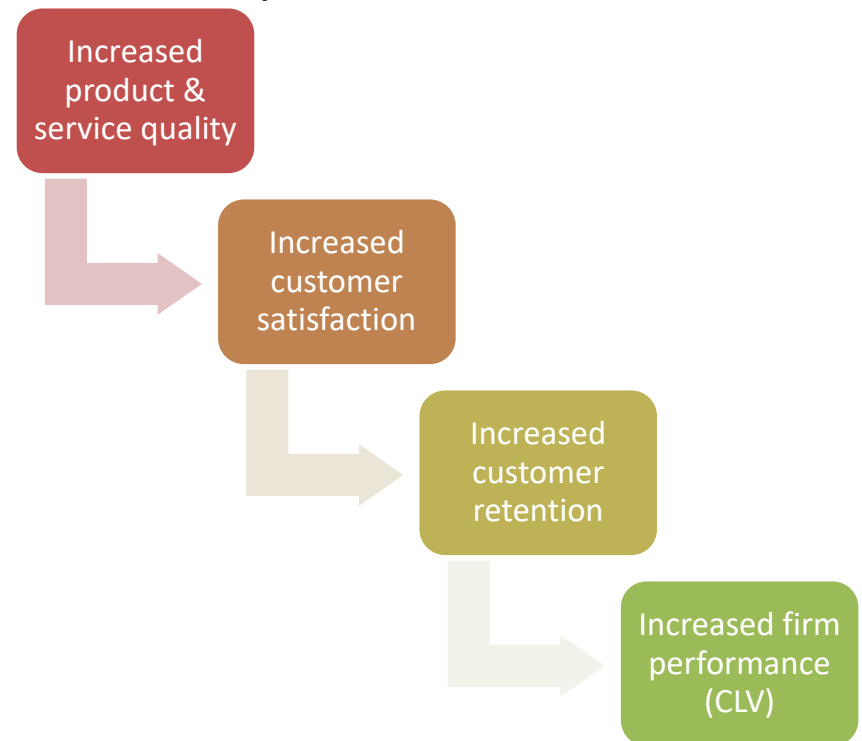


“5% improvement in customer retention can cause an increase in profitability between 25% and 85%....”

Introduction

- Two main research streams in customer retention:
 1. Effects of marketing variables on customer retention
 2. Building statistical models to estimate/predict customer retention decisions
- Integrated relationships addressed in customer retention are described in **Figure 4.1** p. 64

Common thread in these relationships:



Introduction

Stream 1 example:

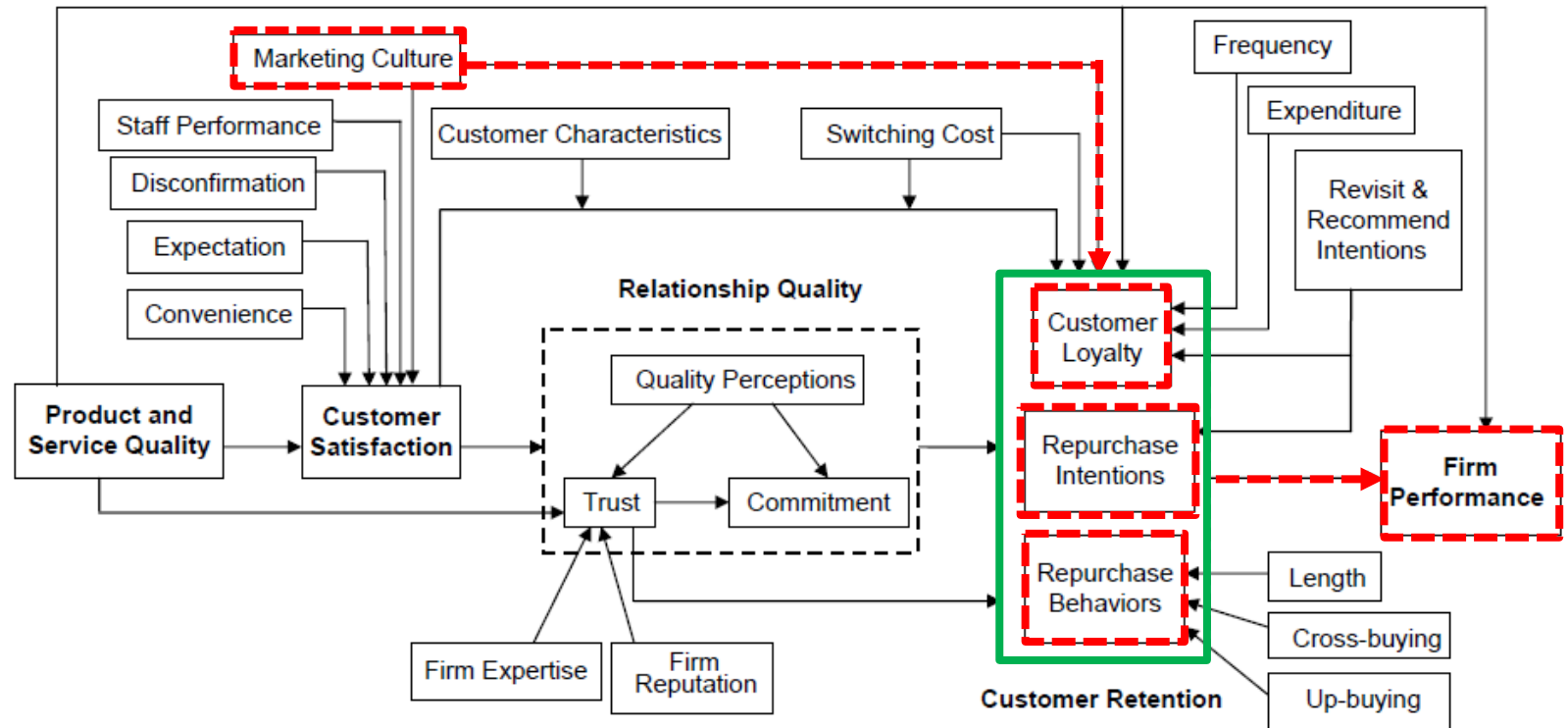


Figure 4.1 Integrated relationships addressed in customer retention.

Introduction

Stream 2 example:

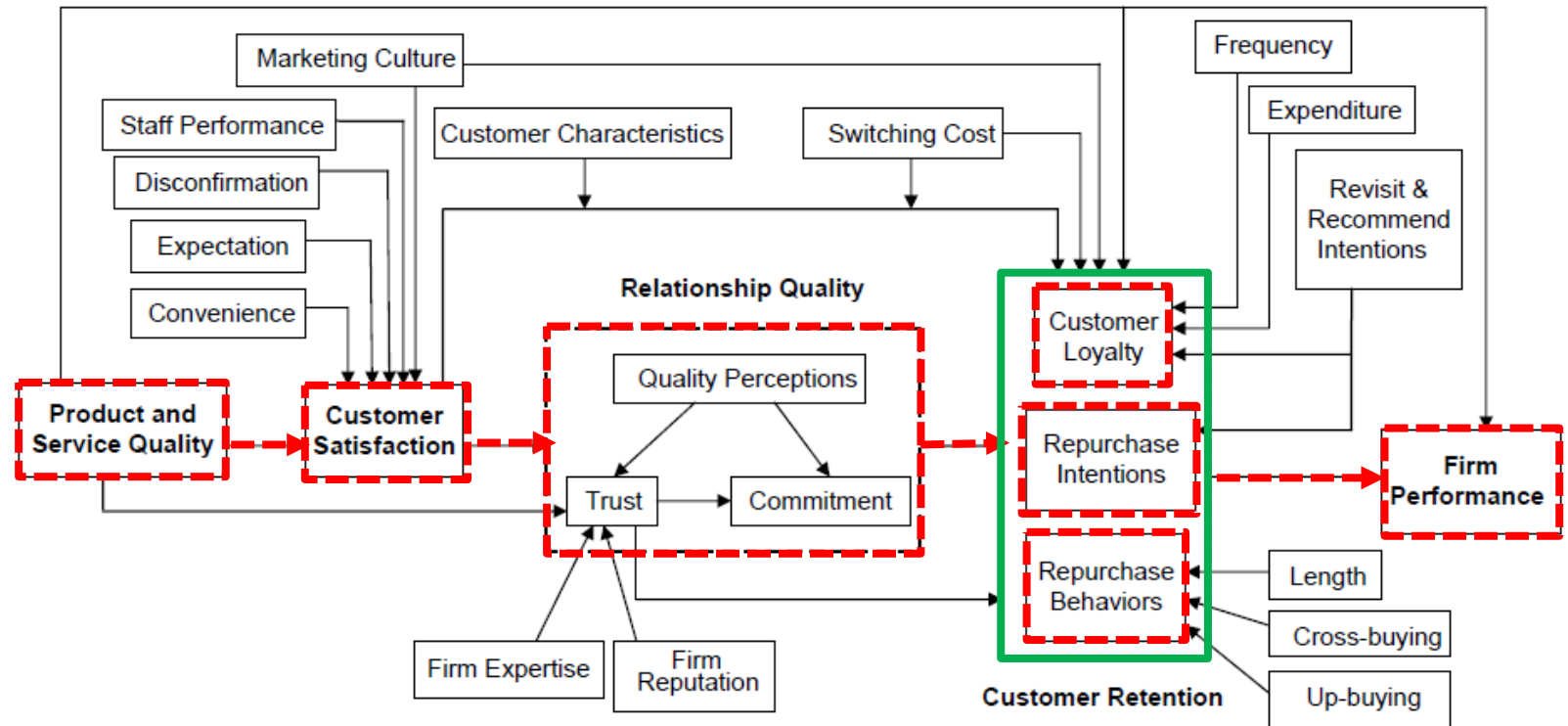


Figure 4.1 Integrated relationships addressed in customer retention.

Introduction

- Some **key questions** often of interest once customers have been acquired:

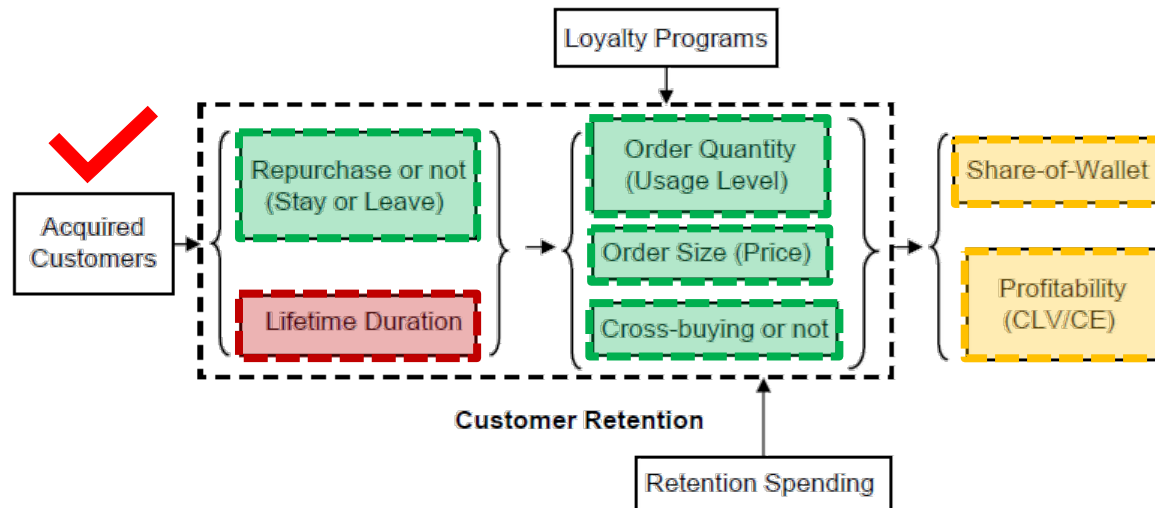


Figure 4.2 Issues addressed in customer retention modeling.

- First question: **contractual** or **non-contractual**

Demo Data

- **B2C** customer information (2 data tables) on **500 customers** who made their first purchase in quarter 1
- **Table 1**: each customer's **transactional data** over 12 quarters (6000 rows, 7 variables)

Variable	
<i>Customer</i>	<i>Customer number (from 1 to 500)</i>
<i>Quarter</i>	<i>The quarter (from 1 to 12) when the transactions occurred</i>
<i>Purchase</i>	<i>1 when the customer purchased in the given quarter, 0 if no purchase occurred in that quarter</i>
<i>Order_Quantity</i>	<i>The dollar value of the purchases in the given quarter</i>
<i>Crossbuy</i>	<i>The number of different categories purchased in a given quarter</i>
<i>Ret_Expense</i>	<i>Dollars spent on marketing efforts to try and retain that customer in the given quarter</i>
<i>Ret_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and retain that customer in the given quarter</i>

Demo Data

- **Table 2:** each customer's demographic information (500 rows, 8 variables)

Variable	
<i>Customer</i>	<i>Customer number (from 1 to 500)</i>
<i>Gender</i>	<i>1 if the customer is male, 0 if the customer is female</i>
<i>Married</i>	<i>1 if the customer is married, 0 if the customer is not married</i>
<i>Income</i>	<i>1 if income < \$30 000</i>
	<i>2 if \$30 001 < income < \$45 000</i>
	<i>3 if \$45 001 < income < \$60 000</i>
	<i>4 if \$60 001 < income < \$75 000</i>
	<i>5 if \$75 001 < income < \$90 000</i>
	<i>6 if income > \$90 001</i>
<i>First_Purchase</i>	<i>The value of the first purchase made by the customer in quarter 1</i>
<i>Loyalty</i>	<i>1 if the customer is a member of the loyalty program, 0 if not</i>
<i>Share-of-Wallet (SOW)</i>	<i>The percentage of purchases the customer makes from the given firm given the total amount of purchases across all firms in that category</i>
<i>CLV</i>	<i>The discounted value of all expected future profits, or customer lifetime value</i>

- **Merge** demographic data with transaction data
- **Note:** SAS code & output provided at the end of Chapter 4

Demo: Import Data Tables

```
/* Import Data */
```

```
proc import out = customer_trans  
    datafile = "Your path\Chapter 4 - Customer Retention Data -  
transactions.xls" dbms = excel replace;  
    getnames = yes;  
    mixed = no;  
    scantext = yes;  
    usedate = yes;  
    scantime = yes;  
  
run;
```

```
proc import out = customer_demo  
    datafile = "Your path\Chapter 4 - Customer Retention Data -  
demographics.xls" dbms = excel replace;  
    getnames = yes;  
    mixed = no;  
    scantext = yes;  
    usedate = yes;  
    scantime = yes;  
  
run;
```

Demo: Merge Data Tables

```
/* Merge Data */
```

```
proc sql;
```

```
    create table customer_retention as
```

```
    select distinct a.*, b.*
```

```
    from customer_trans a left join  
customer_demo b
```

```
    on a.customer = b.customer order by  
customer;
```

```
quit;
```

Feature Engineering

- **Engineer** the following features to be used in the models (per customer):

Feature	Definition
Lag purchase	$lag(purchase)$
Lag cross-buy	$lag(crossbuy)$
Average order quantity	For <u>quarter q</u> : $\frac{\sum_{i=1}^q order_quantity_i}{q}$
Binary cross-buy	$CB = \begin{cases} 1, & Crossbuy > 1 \\ 0, & otherwise \end{cases}$

Feature Engineering: Data Snapshot

Customer	Quarter	Purchase	LagPurchase (Incorrect)	LagPurchase (Correct)
1	12	0	0	0
2	1	1	0	
2	2	0	1	1
2	3	0	0	0
2	4	0	0	0
2	5	0	0	0
2	6	0	0	0
2	7	0	0	0
2	8	0	0	0
2	9	0	0	0
2	10	0	0	0
2	11	0	0	0
2	12	1	0	0
3	1	1	1	
3	2	0	1	1
3	3	0	0	0
3	4	0	0	0
3	5	0	0	0
3	6	1	0	0
3	7	0	1	1
3	8	0	0	0
3	9	0	0	0
3	10	0	0	0
3	11	0	0	0
3	12	0	0	0

Engineering carried out **per customer**:

- Lagged features
 - If lag applied to entire variable ignoring customer segments, first lagged value per customer will be the Q12 value of the previous customer
 - **To avoid**: set first lagged value of each customer to missing

Demo: Feature Engineering

```
/* Feature Engineering */
```

```
data retention_dat;  
set customer_retention;  
by customer; /* Do engineering by customer group */
```

```
Lag_purchase = lag(purchase);  
if first.customer then do; lag_purchase = .; end;
```

```
Lag_crossbuy = lag(crossbuy);  
if first.customer then do; lag_crossbuy = .; end;
```

```
if first.customer then do; quantity_sum = order_quantity; end;  
else do; quantity_sum + order_quantity; end;  
avg_order_quantity = quantity_sum / quarter;
```

```
if crossbuy > 1 then crossbuy_bin = 1; else crossbuy_bin = 0;
```

```
run; quit;
```


REPURCHASE OR NOT?

Drivers of repurchase
behaviour

Nr of repeat purchases

Predictive accuracy



Repurchase or Not

- A **big concern** is that significant resources spent on acquiring a customer will go to waste if the customer doesn't repurchase, **after their first purchase**, for an extended period of time

$$Repurchase = \begin{cases} 1, & \text{repurchase occurs} \\ 0, & \text{otherwise} \end{cases}$$

- **Classification problem**, different models to consider:
 - Logistic regression (**logit** or probit **link**)
 - Decision tree
 - Neural network
 - etc.

Repurchase or Not: Example p. 73

Empirical example p. 73

- Objectives
 - Determine **drivers** of customer repurchase behaviour
 - Interpret parameter estimates
 - Predict** expected number of repeat purchases per customer
 - Determine model's **predictive accuracy**

Information required for this model:

Dependent variable

Purchase 1 when the customer purchased in the given quarter, 0 if no purchase occurred in that quarter

Independent variables

Lag_Purchase 1 if the customer purchased in the previous quarter, 0 if no purchase occurred in the previous quarter

Avg_Order_Quantity The average dollar value of the purchases in all previous quarters
Ret_Expense Dollars spent on marketing efforts to try and retain that customer in the given quarter

Ret_Expense_SQ Square of dollars spent on marketing efforts to try and retain that customer in the given quarter

Gender 1 if the customer is male, 0 if the customer is female

Married 1 if the customer is married, 0 if the customer is not married

Income 1 if income < \$30 000
2 if \$30 001 < income < \$45 000
3 if \$45 001 < income < \$60 000
4 if \$60 001 < income < \$75 000
5 if \$75 001 < income < \$90 000
6 if income > \$90 000

First_Purchase The value of the first purchase made by the customer in quarter 1

Loyalty 1 if the customer is a member of the loyalty program, 0 if not

- Fit a **logistic regression**
- Calculate and interpret the **odds ratios**
- Apply **decision rule** to obtain class predictions
- Calculate the **confusion matrix** and use it to assess **predictive accuracy**

Repurchase or Not: Variable Type Error



Categorical variable types were not set correctly



Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-4.9881	0.2244	493.9003	<.0001
lpurchase	1	2.1446	0.0996	463.6408	<.0001
avg_order_quantity	1	0.0184	0.000660	775.1934	<.0001
ret_expense	1	0.1071	0.00975	120.6179	<.0001
ret_expense_sq	1	-0.00237	0.000213	123.9991	<.0001
gender	1	-0.2023	0.0931	4.7182	0.0298
married	1	-0.0589	0.1084	0.2949	0.5871
income	1	0.1317	0.0318	17.1545	<.0001
first_purchase	1	0.00122	0.00101	1.4693	0.2255
loyalty	1	0.2679	0.1003	7.1384	0.0075



Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-5.5806	0.2633	449.0633	<.0001
Lag_purchase	1	1	3.1410	0.1246	635.9680	<.0001
avg_order_quantity		1	0.0153	0.000703	476.3334	<.0001
Ret_Expense		1	0.1290	0.0110	136.2563	<.0001
Ret_Expense_SQ		1	-0.00280	0.000236	140.8474	<.0001
Gender	1	1	-0.1692	0.0995	2.8906	0.0891
Married	1	1	-0.0217	0.1161	0.0348	0.8521
Income	2	1	0.3702	0.2008	3.3988	0.0652
Income	3	1	0.4341	0.2015	4.6402	0.0312
Income	4	1	0.6734	0.1985	11.5026	0.0007
Income	5	1	0.8431	0.2016	17.4904	<.0001
Income	6	1	0.7771	0.2330	11.1203	0.0009
First_Purchase		1	0.00130	0.00108	1.4452	0.2293
Loyalty	1	1	0.2596	0.1079	5.7825	0.0162

Repurchase or Not: Significant Covariates

Drivers of repurchase behaviour

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-5.5806	0.2633	449.0633	<.0001
Lag_purchase	1	1	3.1410	0.1246	635.9680	<.0001
avg_order_quantity		1	0.0153	0.000703	476.3334	<.0001
Ret_Expense		1	0.1290	0.0110	136.2563	<.0001
Ret_Expense_SQ		1	-0.00280	0.000236	140.8474	<.0001
Gender	1	1	-0.1692	0.0995	2.8906	0.0891
Married	1	1	-0.0217	0.1161	0.0348	0.8521
Income	2	1	0.3702	0.2008	3.3988	0.0652
Income	3	1	0.4341	0.2015	4.6402	0.0312
Income	4	1	0.6734	0.1985	11.5026	0.0007
Income	5	1	0.8431	0.2016	17.4904	<.0001
Income	6	1	0.7771	0.2330	11.1203	0.0009
First_Purchase		1	0.00130	0.00108	1.4452	0.2293
Loyalty	1	1	0.2596	0.1079	5.7825	0.0162

- The following are not drivers of repurchase behaviour:
 - Gender
 - Married
 - Income 2
 - First_purchase
- We need the odds ratios to understand how changes in the drivers (significant predictors) could change the response probability

Repurchase or Not: Odds Ratios

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Lag_purchase 1 vs 0	23.126	18.117	29.521
avg_order_quantity	1.015	1.014	1.017
Ret_Expense	1.138	1.113	1.163
Ret_Expense_SQ	0.997	0.997	0.998
Gender 1 vs 0	0.844	0.695	1.026
Married 1 vs 0	0.979	0.779	1.229
Income 2 vs 1	1.448	0.977	2.147
Income 3 vs 1	1.544	1.040	2.291
Income 4 vs 1	1.961	1.329	2.894
Income 5 vs 1	2.324	1.565	3.450
Income 6 vs 1	2.175	1.378	3.434
First_Purchase	1.001	0.999	1.003
Loyalty 1 vs 0	1.296	1.049	1.602

- **Lag purchase**: odds of repurchase increase by $(23.126-1)*100 > 2000\%$ for customers that purchased in previous quarter compared to customers that didn't
- **Income 5**: odds of repurchase increase by $(2.324-1)*100 = 132.4\%$ for customers in income group 5 compared to customers in income group 1
- **Question**: *Ret_Exp* OR?

Repurchase or Not: Retention Expense Odds Ratio

Using the logit:

$$\hat{p}'_i = \ln\left(\frac{\hat{p}_i}{1 - \hat{p}_i}\right) = \ln odds_i = -3.4595 + 0.1290x_i - 0.0028(x_i)^2,$$

and

$$\hat{p}'_{i+1} = \ln\left(\frac{\hat{p}_{i+1}}{1 - \hat{p}_{i+1}}\right) = \ln odds_{i+1} = -3.4595 + 0.1290(x_i + 1) - 0.0028(x_i + 1)^2.$$

The difference is:

$$\begin{aligned} & \ln odds_{i+1} - \ln odds_i \\ &= \ln\left(\frac{odds_{i+1}}{odds_i}\right) \\ &= 0.1262 - 0.0056x_i \end{aligned}$$

Hence, the Odds Ratio is:

$$OR = \left(\frac{odds_{i+1}}{odds_i}\right) = \exp(0.1262 - 0.0056 x_i)$$

$$\underline{OR} = \exp(0.1262 - 0.0056 \cdot 10) = 1.073$$

⇒ Odds of repurchase increase by 7.3% when retention expense increases from \$10 to \$11

Repurchase or Not: Model Assessment

The SAS System

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of pred_pur by act_pur			
	pred_pur	act_pur		
		0	1	Total
0	2504 45.53 92.47 88.73	204	2708	
		3.71	49.24	
		7.53		
		7.62		
1	318 5.78 11.39 11.27	2474	2792	
		44.98	50.76	
		88.61		
		92.38		
Total	2822 51.31	2678 48.69	5500 100.00	

- Confusion matrix

- Only quarter > 1 (why?)

- Total 5500 not 6000?

- Misclassification rate

$$\frac{204 + 318}{5500} \times 100 = 9.49\%$$

ORDER QUANTITY

Drivers of order
quantity (value)

Expected order
quantity

Predictive accuracy



Order Quantity



- Not sufficient to just focus on repurchasing
- Emphasis should also be placed on how much value such repurchases could provide
- **Order quantity (value)** an indicator of customer's future value to the firm
 - Justification for retention expense

Order Quantity: Data

Empirical example p. 87

- Objectives
 - Determine drivers of order quantity (value)
 - Predict expected order quantity per customer
 - Determine model's predictive accuracy

Information required for this model:

Dependent variables

<i>Purchase</i>	1 when the customer purchased in the given quarter, 0 if no purchase occurred in that quarter
<i>Order_Quantity</i>	The dollar value of the purchases in the given quarter

Independent variables

<i>Lambda (λ)</i>	The computed inverse Mills ratio from the acquisition model
<i>Lag_Purchase</i>	1 if the customer purchased in the previous quarter, 0 if no purchase occurred in the previous quarter
<i>Avg_Order_Quantity</i>	The average dollar value of the purchases in all previous quarters
<i>Ret_Expense</i>	Dollars spent on marketing efforts to try and retain that customer in the given quarter
<i>Ret_Expense_SQ</i>	Square of dollars spent on marketing efforts to try and retain that customer in the given quarter
<i>Gender</i>	1 if the customer is male, 0 if the customer is female
<i>Married</i>	1 if the customer is married, 0 if the customer is not married
<i>Income</i>	1 if income < \$30 000 2 if \$30 001 < income < \$45 000 3 if \$45 001 < income < \$60 000 4 if \$60 001 < income < \$75 000 5 if \$75 001 < income < \$90 000 6 if income > \$90 001
<i>First_Purchase</i>	The value of the first purchase made by the customer in quarter 1
<i>Loyalty</i>	1 if the customer is a member of the loyalty program, 0 if not

!! Two targets

Order Quantity: Two-stage Model

$$E(\text{Order quantity}) = P(\text{Purchase} = 1)E(\text{Order quantity}|\text{Purchase} = 1)$$

where

- **Purchase** is modelled using a **probit** (**why?**) model
- **Order quantity** is modelled using **OLS**

Hence

$$\begin{aligned} E(\widehat{\text{Order quantity}}) &= P(\text{Purchase} = 1)E(\text{Order quantity}|\text{Purchase} = 1) \\ &= \Phi\left(\frac{X\hat{\beta}}{\sigma}\right) \left[Z\hat{\alpha} + \mu \lambda\left(\frac{X\hat{\beta}}{\sigma}\right) \right] \end{aligned}$$

Inverse Mills Ratio (IMR) **why?**

where

$$\lambda\left(\frac{X\hat{\beta}}{\sigma}\right) = \frac{\phi\left(\frac{X\hat{\beta}}{\sigma}\right)}{\Phi\left(\frac{X\hat{\beta}}{\sigma}\right)}$$

Order Quantity: Example p. 87

- Use **two-stage modelling** to predict order quantity
- **SAS code** (**explained in Acquisition Part 2 slides**)
 - PROC LOGISTIC (link = probit)
 - Use a DATA STEP to calculate the INVERSE MILLS RATIO
 - PROC REG/GLM including IMR as predictor
 - Use a DATA STEP to apply prediction calculation formula and PROC SQL to calculate MAD
- **Results**
 - Mean order quantity (**where Purchase=1 & Quarter > 1**) = 265.14
 - MAD = 60.47
 - Random MAD = 71.53

Order Quantity Stage 1: Drivers of Purchase Probability

Stage 1: purchase probability

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-3.0259	0.1351	501.7454	<.0001
Lag_purchase	1	1	1.7553	0.0658	711.0886	<.0001
avg_order_quantity		1	0.00821	0.000373	483.5287	<.0001
Ret_Expense		1	0.0682	0.00564	146.3325	<.0001
Ret_Expense_SQ		1	-0.00149	0.000122	149.9519	<.0001
Gender	1	1	-0.0902	0.0525	2.9486	0.0860
Married	1	1	-0.0102	0.0609	0.0279	0.8673
Income	2	1	0.2297	0.1079	4.5312	0.0333
Income	3	1	0.2732	0.1080	6.3998	0.0114
Income	4	1	0.3810	0.1064	12.8294	0.0003
Income	5	1	0.4778	0.1080	19.5815	<.0001
Income	6	1	0.4507	0.1231	13.4064	0.0003
First_Purchase		1	0.000789	0.000569	1.9231	0.1655
Loyalty	1	1	0.1464	0.0572	6.5526	0.0105

Significant predictors of repurchase highlighted in green

Order Quantity Stage 2: Drivers of Order Quantity

Stage 2: order quantity

Parameter	Estimate		Standard Error	t Value	Pr > t
Intercept	-278.7217619	B	30.13759331	-9.25	<.0001
Lag_purchase 1	204.2527093	B	17.46511980	11.69	<.0001
Lag_purchase 0	0.0000000	B	.	.	.
avg_order_quantity	0.9293741		0.03672688	25.31	<.0001
Ret_Expense	3.0523661		0.34672580	8.80	<.0001
Ret_Expense_SQ	-0.0742273		0.00776505	-9.56	<.0001
Gender 1	10.3091304	B	2.68817766	3.83	0.0001
Gender 0	0.0000000	B	.	.	.
Married 1	-2.2915934	B	3.00872729	-0.76	0.4463
Married 0	0.0000000	B	.	.	.
Income 2	20.8440337	B	7.29738365	2.86	0.0043
Income 3	29.4581029	B	7.17887564	4.10	<.0001
Income 4	45.3992483	B	7.06997322	6.42	<.0001
Income 5	58.9209495	B	7.13762364	8.25	<.0001
Income 6	70.4027030	B	7.41941304	9.49	<.0001
Income 1	0.0000000	B	.	.	.
First_Purchase	0.2642990		0.02850196	9.27	<.0001
Loyalty 1	9.8903666	B	3.00741690	3.29	0.0010
Loyalty 0	0.0000000	B	.	.	.
imr_purchase	112.3896762		12.08638597	9.30	<.0001

- **IMR** significant
 - Error terms are correlated
- **Drivers** of order quantity:
 - Lag_purchase
 - Average order quantity
 - Retention expense
 - Gender
 - Income
 - First purchase amount
 - Loyalty member

Order Quantity Stage 2: Retention Expense

- Keeping all other covariates constant:
 - At $X = x$: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$ ----- ①
 - At $X = x + 1$: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(x + 1) + \hat{\beta}_2(x + 1)^2$ ----- ②
- For this example:
 - At $X = x$: $\hat{y} = -278.72 + 3.05x - 0.07x^2$ ----- ①
 - At $X = x + 1$: $\hat{y} = -278.72 + 3.05(x + 1) - 0.07(x + 1)^2$ ----- ②
- The effect of a 1 unit change in X (Retention Expense) on $\hat{y}$:
$$\text{②} - \text{①}$$
$$= 2.98 - 0.14x$$

The effect of a 1 unit change in Retention Expense on $\hat{y}$ depends on level of Retention Expense

⇒ Order quantity increases by \$1.44 when retention expense increases **from \$10 to \$11**

Order Quantity: Model Assessment

Predictive accuracy (quarter > 1)

mad	mape	random_mad	random_mape
60.47293	0.244896	71.53183	0.328151

- **Two-stage model:**
 - **MAD:** 60.47
 - Each prediction deviates from observed value by \$60.47
 - **MAPE:** 0.245
 - Each prediction deviates from observed value by 24.5%
- **Random guess** model:
 - MAD: 71.53
 - MAPE: 0.328
- Two-stage model **better** than random guess

CROSS-BUYING

Drivers of cross-buying
behaviour

Probability of cross-
buying

Predictive accuracy



Cross-Selling



Most common technique
used by companies to
increase:

repeat purchase incidence
quantity
revenues



Assumption: loyal customers are more likely to
cross-buy from the same company



A **key question** then: can we determine which
customers have the **highest likelihood of cross-
buying** in multiple categories?

Cross-Buying: Example p. 92

Empirical example p. 92

- Objectives
 1. Identify the drivers of customer cross-buying behaviour
 2. Interpret parameter estimates from cross-buying model
 3. Predict whether a customer is likely to cross-buy or not
 4. Determine model's predictive accuracy

Information required for this model:

Dependent variable	
Crossbuy	The number of product categories a customer purchased during a given quarter
Independent variables	
Lag_Purchase	1 if the customer purchased in the previous quarter, 0 if no purchase occurred in the previous quarter
Lag_Crossbuy	The number of product categories a customer purchased during the previous quarter
Avg_Order_Quantity	The average dollar value of the purchases in all previous quarters
Ret_Expense	Dollars spent on marketing efforts to try and retain that customer in the given quarter
Ret_Expense_SQ	Square of dollars spent on marketing efforts to try and retain that customer in the given quarter
Gender	1 if the customer is male, 0 if the customer is female
Married	1 if the customer is married, 0 if the customer is not married
Income	1 if income < \$30 000 2 if \$30 001 < income < \$45 000 3 if \$45 001 < income < \$60 000 4 if \$60 001 < income < \$75 000 5 if \$75 001 < income < \$90 000 6 if income > \$90 001
First_Purchase	The value of the first purchase made by the customer in quarter 1
Loyalty	1 if the customer is a member of the loyalty program, 0 if not

Created cross-buy indicator (see slide 16)

Lag_purchase, Lag_crossbuy and Avg_order_qty also slide 16

Will be dependent variable here

Cross-Buying: Example p. 92

- **SAS code**
 - Predict **cross-buy probability** using PROC LOGISTIC with LINK=LOGIT **where Purchase=1** (probability of cross-buy given that a purchase has occurred)
 - Use a DATA STEP (cut-off = 0.5) to predict **nr of customers that will cross-buy**
 - Use PROC FREQ to calculate **confusion matrix**
 - Determine **predictive accuracy (quarter > 1)**

Cross-Buying: Significant Covariates

Drivers of cross-buying behaviour

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-8.7890	0.5312	273.7725	<.0001
Lag_purchase	1	1	1.6980	0.3962	18.3690	<.0001
Lag_crossbuy		1	0.2050	0.0434	22.2802	<.0001
avg_order_quantity		1	0.0101	0.000880	131.3291	<.0001
Ret_Expense		1	0.0435	0.00985	19.4673	<.0001
Ret_Expense_SQ		1	-0.00088	0.000221	15.8978	<.0001
Gender	1	1	0.2920	0.0990	8.7030	0.0032
Married	1	1	0.1890	0.1119	2.8518	0.0913
Income	2	1	0.2656	0.2744	0.9365	0.3332
Income	3	1	0.4959	0.2683	3.4158	0.0646
Income	4	1	0.9096	0.2642	11.8534	0.0006
Income	5	1	1.0929	0.2667	16.7876	<.0001
Income	6	1	1.1704	0.2773	17.8180	<.0001
First_Purchase		1	0.0242	0.00126	372.5639	<.0001
Loyalty	1	1	0.2873	0.1094	6.8914	0.0087

Not significant

Cross-Buying: Odds Ratios



Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Lag_purchase 1 vs 0	5.463	2.513	11.876
Lag_crossbuy	1.227	1.127	1.337
avg_order_quantity	1.010	1.008	1.012
Ret_Expense	1.044	1.024	1.065
Ret_Expense_SQ	0.999	0.999	1.000
Gender 1 vs 0	1.339	1.103	1.626
Married 1 vs 0	1.200	0.970	1.504
Income 2 vs 1	1.304	0.762	2.233
Income 3 vs 1	1.642	0.970	2.778
Income 4 vs 1	2.483	1.480	4.168
Income 5 vs 1	2.983	1.768	5.032
Income 6 vs 1	3.223	1.872	5.551
First_Purchase	1.025	1.022	1.027
Loyalty 1 vs 0	1.333	1.076	1.652

- **Effect** of drivers of cross-buying behaviour:
 - **Lag_crossbuy**: odds of CB increase by 22.7% if nr of categories cross-bought (previous quarter) increases by 1
 - **Income 6**: odds of CB increase by >200% if customer is in income level 6 compared to income level 1
 - **Loyalty**: odds of CB increase by 33.3% if customer is part of loyalty programme compared to customer that isn't

Cross-Buying: Model Assessment

The SAS System

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of pred_cb by act_cb			
	pred_cb	act_cb		
		0	1	Total
0	0	984	292	1276
		36.74	10.90	47.65
		77.12	22.88	
		74.72	21.45	
1	1	333	1069	1402
		12.43	39.92	52.35
		23.75	76.25	
		25.28	78.55	
Total	Total	1317	1361	2678
		49.18	50.82	100.00

Binary cross-buy distribution:

crossbuy_bin	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	1665	52.39	1665	52.39
1	1513	47.61	3178	100.00

➤ 0.5 cut-off appropriate?

• Predictive accuracy

➤ Only quarter > 1 (why?)

➤ Total 2678 not 6000?

• Misclassification rate

$$\frac{292 + 333}{2678} \times 100 = 23.34\%$$

Summary

Customers have been acquired

Focus: customer retention

- Will the customer **repurchase** after first purchase?
 - Repurchase probability prediction and drivers
- Spend some money to get the customers to repurchase
 - Can we **justify** the retention **expense**?
 - What does the **future value** of the customer to the firm look like?
 - Predict order quantity **given** repurchase

Also: customer expansion

- Cross-selling (or **cross-buying**) widely used for retention and expansion
- **Which customers** should be targeted for cross-selling?
- Predict likelihood of cross-buying and identify **characteristics** of such customers

Homework



Classwork (in groups)

Acquisition Part 3

Data: Ch03 acq only

Due: 24 March at 13h00



Prep for 24 March

Kumar & Petersen Sections
3.4, 4.3, 4.6 & 4.7